UNIT 3: Similarity Analysis and Cascade Analysis

Turbo Machinery | Handwritten-style exam notes | 27 Oct 2026

Syllabus covered: Similarity rules, specific speed, Cordier diagram and examples; two-dimensional cascade theory, lift and drag, blade efficiency, loss estimation, compressor and turbine cascades, blade geometry, examples.

1. Why similarity?

Testing full-size machines is expensive. Similarity analysis lets us predict the performance of one machine from another that is geometrically and dynamically similar. Two machines are similar when all their dimensionless groups are the same.

- Geometric similarity: all lengths are in a fixed ratio.
- Kinematic similarity: velocity triangles are similar at corresponding points.
- Dynamic similarity: Reynolds number, Mach number and ratio of specific heats match (for compressible flow).

2. Dimensionless groups

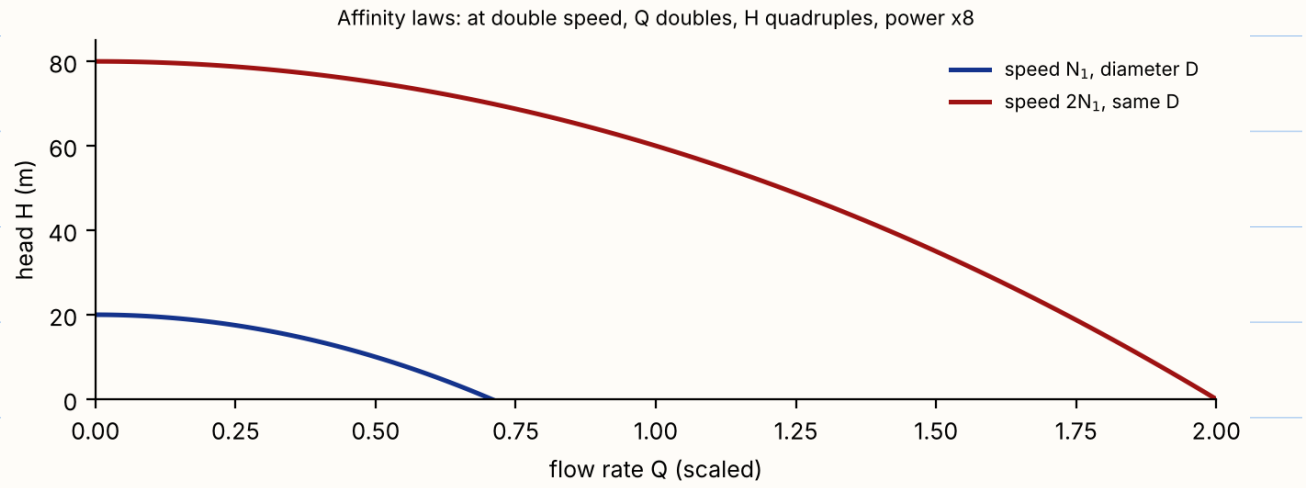
The main groups used in turbomachinery are:

Flow coefficient $\varphi = C_x / U$ Head (stage loading) coefficient $\psi = gH / U^2 = \Delta h_0 / U^2$

Power coefficient $\lambda = P / (\rho N^3 D^5)$ Reynolds number $Re = \rho U c / \mu$ Mach number $M = U/a$

3. Affinity (similarity) laws

Affinity laws: head against flow at two speeds



For the same machine, or two geometrically similar machines, with constant density, the performance changes with speed N and diameter D as:

$$Q_2/Q_1 = (N_2/N_1)(D_2/D_1)^3 \quad H_2/H_1 = (N_2/N_1)^2(D_2/D_1)^2 \quad P_2/P_1 = (N_2/N_1)^3(D_2/D_1)^5$$

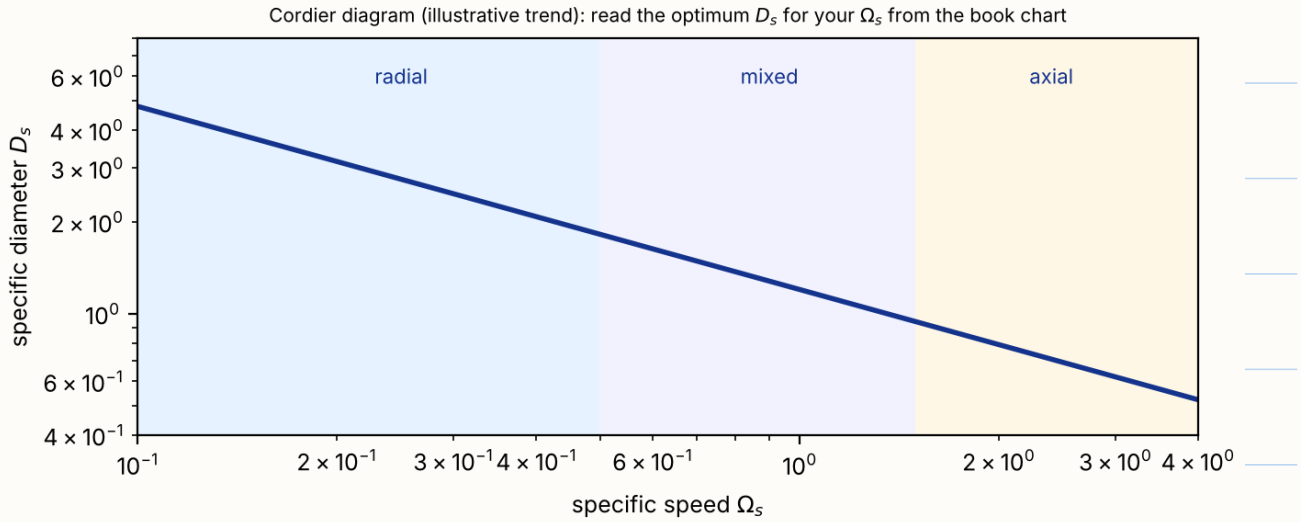
Example: *A pump runs at 1450 rpm and gives a head of 20 m.*

Estimate the head at 2900 rpm with the same diameter.

Solution: $H_2 = 20 \times (2900/1450)^2 = 20 \times 4 = 80$ m.

4. Specific speed

Cordier diagram with radial, mixed and axial zones



Specific speed is a dimensionless parameter that describes the shape of a machine. It depends only on operating point (flow rate and head) and not on size. It is used to choose the machine type.

$$\Omega_s = \omega \sqrt{Q} / (gH)^{3/4} \quad (\omega \text{ in rad/s, } Q \text{ in m}^3/\text{s, } H \text{ in m})$$

Specific diameter is the companion parameter:

$$D_s = D\,(gH)^{1/4} / \sqrt{Q}$$

- Low Ω_s (about 0.1 to 0.5): radial (centrifugal) machine, high head, low flow.
- Medium Ω_s (about 0.5 to 1.5): mixed flow machine.
- High Ω_s (above about 1.5): axial flow machine, high flow, low head.

Example: *A pump has Q = 0.1 m³/s, H = 30 m and N = 1450 rpm. Find the specific speed and the machine type.*

Solution: $\omega = 2\pi \times 1450/60 = 151.9$ rad/s. $\sqrt{Q} = 0.316$. $(gH)^{3/4} = (9.81 \times 30)^{0.75} = 71.0$. So $\Omega_s = 151.9 \times 0.316 / 71.0 = 0.68$. This is a mixed flow pump.

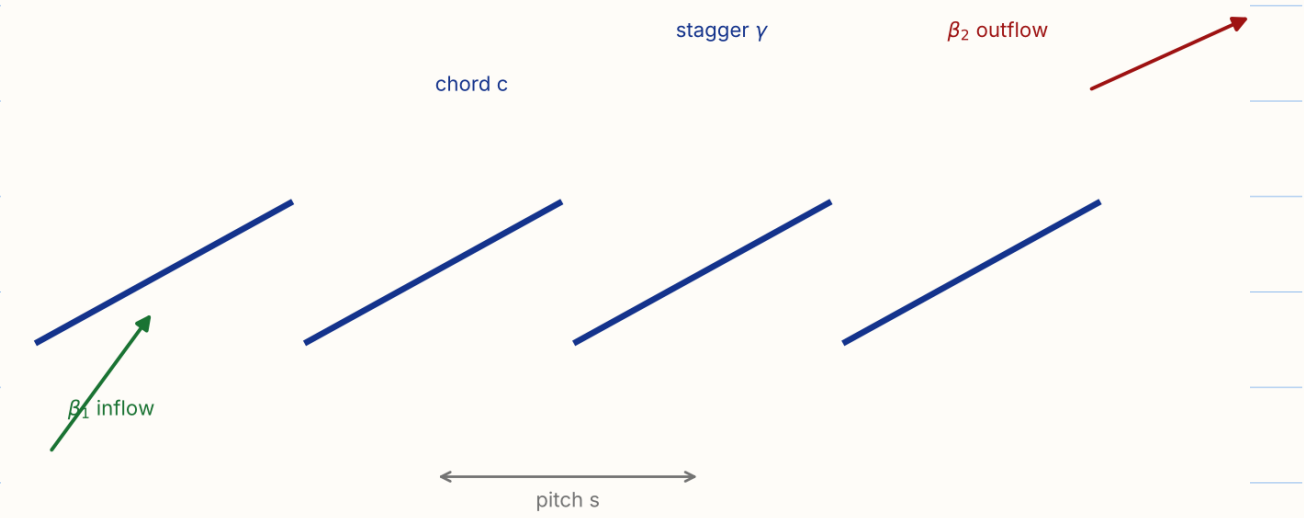
5. Cordier diagram

Cordier plotted the optimum (maximum efficiency) specific diameter D_s against specific speed Ω_s for a large set of real machines. The data collapse onto a single curve. The main message is:

- Low Ω_s needs a large D_s, which is why radial machines have large diameter and small flow.
- High Ω_s needs a small D_s, which is why axial machines are slim and fast.
- The Cordier diagram is used at the preliminary design stage to select the type and approximate size of a machine.

6. Two-dimensional cascade: definitions

Cascade: pitch, chord, stagger and flow angles



A cascade is a row of blades laid out on a plane, as if the blades were unrolled from a stage at a given radius. The main geometric quantities are:

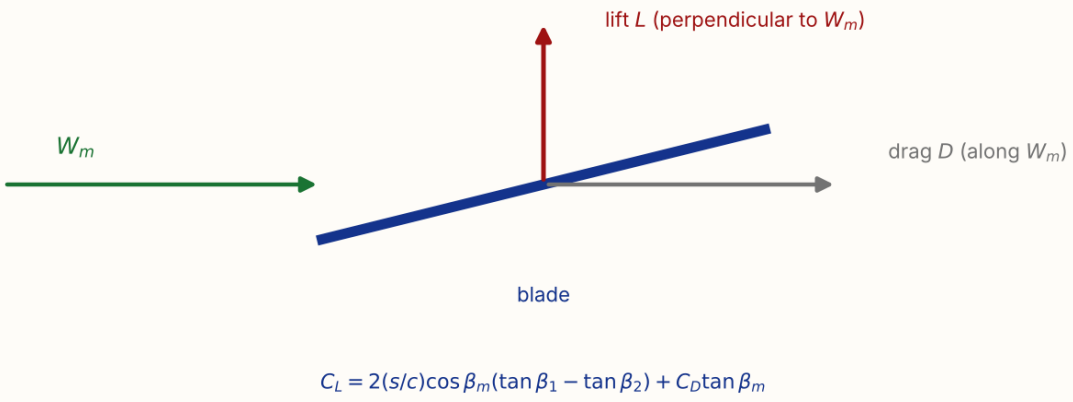
- **Pitch s:** spacing between blades. **Chord c:** length of blade from leading edge to trailing edge.

- **Solidity σ = c / s.** Stagger angle γ: angle of the chord to the axial direction.
- **Inlet flow angle β_1** and **outlet flow angle β_2**: angles of the relative flow measured from axial.
- **Incidence i** = $\beta_1 - \beta_1'$ (difference between actual and blade inlet angle). **Deviation δ** = $\beta_2 - \beta_2'$.

Camber θ = $\beta_1 - \beta_2$ **Mean flow angle tan β_m = ½ (tan β_1 + tan β_2)**

7. Lift and drag in a cascade

Lift and drag on a blade in a cascade



The cascade turns the flow by θ. The change in tangential velocity across the cascade produces a force on the blade. This force can be

split into lift (perpendicular to the mean velocity) and drag (parallel to it).

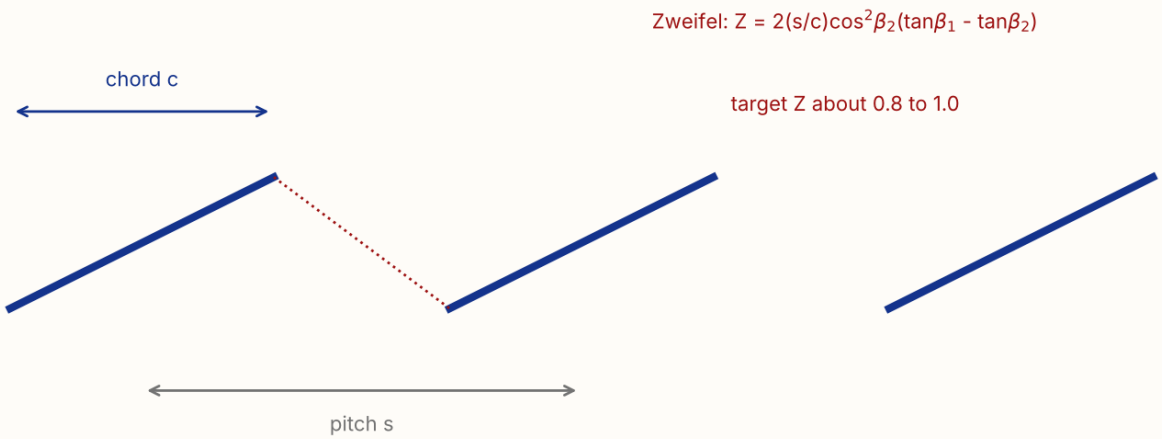
$$C_L = 2\,(s/c)\cos\beta_m\,(\tan\beta_1 - \tan\beta_2) + C_D \tan\beta_m$$

Here C_D is the profile drag coefficient. The first term is the lift that does the turning. The second term is the drag contribution to the force along the axial direction. For good cascades C_D is small compared with C_L, so the lift term dominates.

Lift-to-drag ratio is a measure of cascade quality. Good designs have a high C_L/C_D, which gives high blade efficiency.

8. Zweifel criterion (blade loading)

Zweifel loading: pitch against chord



Zweifel proposed a rule for the best pitch. The loading coefficient should be about 0.8 to 1.0 for minimum loss:

$$Z = 2\,(s/c)\cos^2\beta_2\,(\tan\beta_1 - \tan\beta_2) \quad \text{target } Z \approx 0.8 \text{ to } 1.0$$

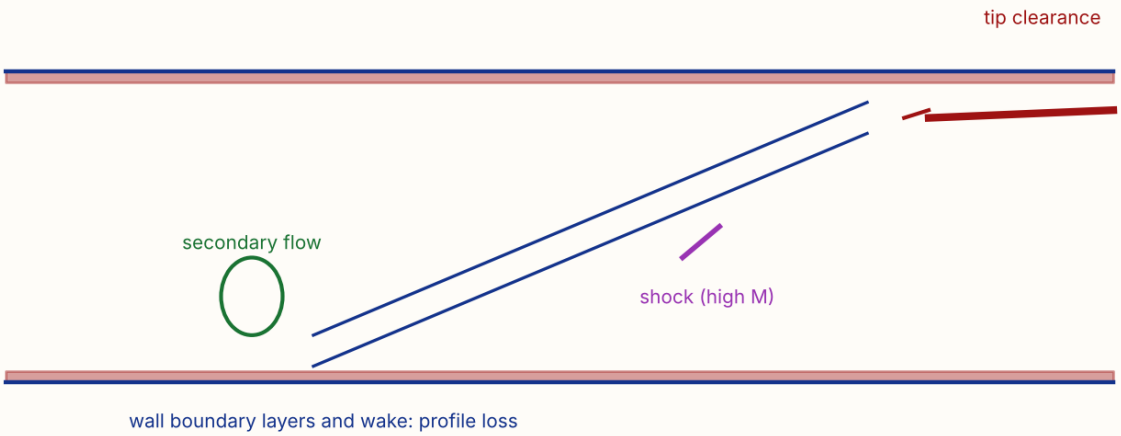
Example: *For a compressor cascade with $\beta_1 = 60°$, $\beta_2 = 30°$ and s/c = 0.6, check the Zweifel number.*

Solution: $\cos^2 30° = 0.75$, $\tan 60° - \tan 30° = 1.732 - 0.577 = 1.155$.

$Z = 2 \times 0.6 \times 0.75 \times 1.155 = 1.04$. This is slightly high, so a slightly larger s/c would reduce loading, at the cost of more blades.

9. Blade efficiency and loss estimation

Cascade passage and loss sources



Losses in a cascade are measured by the pressure loss coefficient:

$$\zeta = (p_{01} - p_{02}) / (\tfrac{1}{2} \rho W_1^2)$$

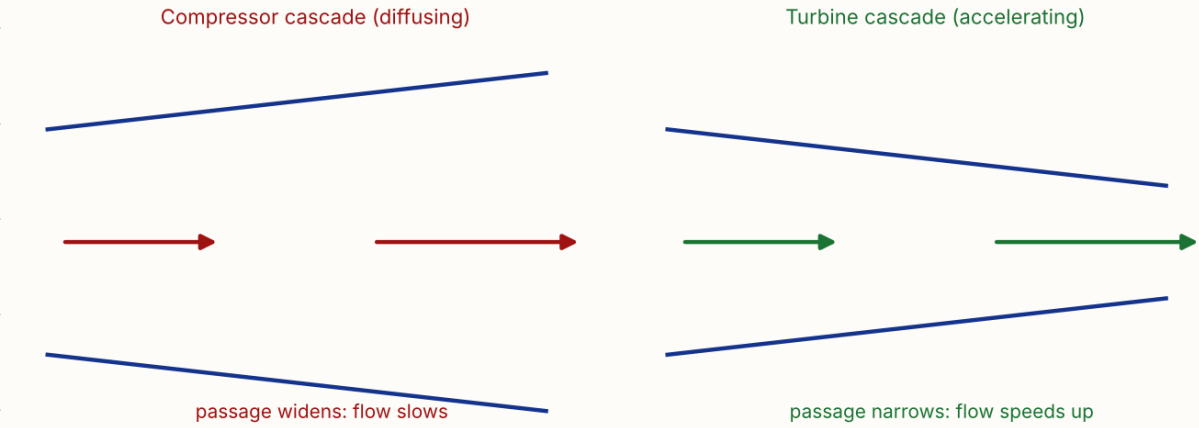
Losses have several sources, and they are added to give the total:

- **Profile loss:** boundary layer growth and wake on the blade surface.
- **Secondary flow loss:** passage vortices and endwall boundary layers.
- **Tip clearance loss:** leakage over the blade tip.
- **Shock loss:** at high Mach number on the blade surface (compressor blades at high relative Mach).

Blade efficiency for a compressor cascade is roughly $1 - (C_D/C_L) \times$ (something of order 1), so high lift and low drag mean high efficiency. Empirical correlations (for example, Ainley and Mathieson) estimate ζ from geometry and flow angle.

10. Compressor and turbine cascades

Diffusing (compressor) vs accelerating (turbine) passage



Item	Compressor cascade	Turbine cascade
Flow	Decelerating (diffusing)	Accelerating (expanding)
Main loss risk	Separation, stall	Mostly profile and trailing-edge loss
Typical loading	Moderate (Zweifel near 0.8–1.0)	Higher turning possible
Key limit	Diffusion factor	Exit Mach and trailing edge thickness

Compressor cascades are limited by diffusion, because the flow must slow down and the boundary layer can separate. Turbine cascades accelerate the flow, so they can be loaded more heavily without separation.

11. Worked example

Example: *Find the mean flow angle for $\beta_1 = 60°$ and $\beta_2 = 30°$.*

Solution: $\tan \beta_m = ½ (1.732 + 0.577) = 1.1547$, so $\beta_m = 49.1°$.

12. Exam quick points

- Similarity: keep φ, ψ, Re, M the same. Affinity: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$.
- Ω_s tells us the machine type. Cordier diagram gives D_s for best efficiency.
- Cascade: s/c, solidity, camber, deviation, incidence. Zweifel target about 0.8 to 1.0.
- Losses: profile, secondary, tip clearance, shock.

13. Practice questions

Q1. Define specific speed and explain how it is used to select a pump type.

Q2. State the affinity laws and use them to find the flow rate when speed is doubled at constant diameter.

Q3. Explain what the Zweifel coefficient represents and what happens if the pitch is too large or too small.